

BHUMIKA AGARWAL

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EXPERIENCE

JPMorgan Chase & Co.

Mumbai, India

Software Engineer II

Oct 2024 – Present

- Owned the **overcharge dispute capability end-to-end** – from requirements through production – designing the **Camunda 7 BPMN** workflow, Java External Task Workers, and Spring Boot services resolving **10k+ dispute cases/month** for credit card customers.
- Modernized inter-service communication across **25+** microservices by building a **SOAP-to-REST translation layer** and onboarding **Kafka SaaS** processing **200k events/sec** at peak – decoupling legacy mainframe tiers from new service tiers.
- Codified microservice infrastructure as code and rolled out **Blue-Green deployment** pipelines across **18+** services and 3 environments, eliminating downtime windows on a regulated dispute platform.
- Diagnosed and remediated a **P1 production incident** in the bonds trading Category Service returning skewed market data, restoring data integrity within **4 hours** under live-market pressure.
- Shipped a macroeconomic event-correlation feature on the React-based bonds trading platform, surfacing real-time linkage between global events and **100+** securities for desk traders.
- Hardened **30+** REST endpoints with **Spring Security and JWT** and authored JUnit suites lifting code coverage past **70%**, stabilizing CI signal across the bonds platform.

Cognizant

Mumbai, India

Software Engineer

Jul 2022 – Jul 2024

- Built **Spring Boot microservices and REST APIs** powering high-volume BFSI transaction workflows for enterprise banking clients.
- Tuned Hibernate/JPA query patterns and SQL execution plans across **[X]** endpoints, **cutting average API latency by 30%** and reducing peak-hour database contention.
- Delivered reusable ReactJS dashboard components wired to backend services; drove JUnit and Mockito test rigor across service and component layers.

PROJECTS

Regulatory Approval System | *Java 17, Spring Boot 3, Camunda 7, Spring Security, JWT, Docker* | [Link](#)

- Engineered a BPMN-based regulatory approval platform on Camunda 7 using the **External Task Worker pattern** – 5 specialized workers handling risk scoring, compliance, escalation, notification, and completion – with **SLA enforcement (8h–48h)** and exponential-backoff retry.
- Synchronized JWT and **6-role RBAC** between Spring Security and Camunda Identity Service, enabling auditable workflow governance and enterprise-grade approval chains.

UrbanNexus | *Java 17, Spring Boot 3.5, React 19, TypeScript, MySQL, JWT, Docker* | [Link](#)

- Architected a multi-portal residential operations platform (Resident, Technician, SuperAdmin) with **role-isolated JWT authentication** across payment, booking, and operational workflows.
- Designed a transactional booking engine with automated technician matching, payment lifecycle orchestration, and JPA **@PrePersist** hooks for tax and audit-field automation.
- Shipped **25+** REST endpoints across a Spring Boot 3.5 + React 19 + TypeScript stack with audit trails, overdue-payment automation, and dual-mode H2/MySQL deployment.

TECHNICAL SKILLS

Languages: Java, TypeScript, JavaScript, SQL

Backend: Spring Boot, Spring Security, Spring Data JPA, Hibernate, REST APIs, Microservices, Camunda 7 (BPMN), JWT, RBAC

Frontend: React.js, TypeScript, Material-UI, HTML5, CSS3

Data & Messaging: MySQL, MongoDB, Apache Kafka, Event-Driven Architecture, Query Optimization

Cloud & DevOps: AWS (ECS, EC2, S3, Route 53), Docker, Blue-Green Deployment, IaC, GitHub Actions, Jenkins, Maven

Observability & Testing: Splunk, Dynatrace, JUnit 5, Mockito

EDUCATION

University of Petroleum and Energy Studies

Dehradun, Uttarakhand

B.Tech. in Computer Science Engineering

Jun 2018 – Jun 2022